

Unified Cognitive-Engineering Architecture: Knowledge & Memory Subsystem

1. Executive Summary & Root Cause of Inconsistencies

The divergence between **System A (4-Tier)** and **System B (5-Tier)** stems from two architectural conflations:

- Procedural vs. Episodic Conflation (System A, Tier 4):**
 - System A* combined executable logic with telemetry into a single layer (/automation_scripts/).
 - In cognitive architecture, **Procedural Memory** (executable skills/routines) must remain strictly decoupled from **Episodic Memory** (time-indexed trajectory logs and verifier outcomes).
- Metacognitive Indexing vs. Procedural Skills (System A, Tier 3 vs. System B, Tier 5):**
 - System A* merged "The Almanac" (the global index/manifest) with "Skill Definitions" under /inference_skills/.
 - System B* correctly separated the global directory (MAP.md / manifest.jsonl) as Tier 5, but omitted explicit prompt-level skill definitions and episodic run logs.
- Retrieval Caching vs. Conceptual Knowledge (System B, Tier 3):**
 - System B* recognized that human-readable markdown (/wiki_md/) cannot be ingested at high velocity by LLM runtime workers, creating /recall_jsonl/ as an optimized machine-cache layer.

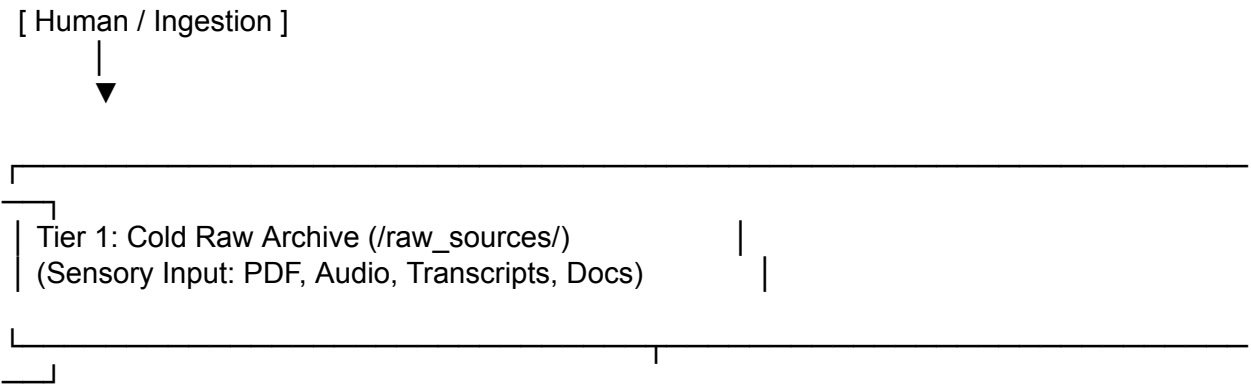
2. Definitive Reconciliation Matrix

UNIFIED 5+1 TIER MEMORY TOPOLOGY			
TIER	CANONICAL DIRECTORY & ARTIFACTS	COGNITIVE MEMORY TYPE	FUNCTION
1	/raw_sources/	Sensory / External Ground Truth	PDFs, Docs, Audio/Vid
2	/master_wiki/	Semantic Memory (Concepts)	Markdown Wiki, Specs
3	/recall_cache/	Working Memory Accelerator	Chunked JSONL, Vectors
4	/skills_and_runtime/	Procedural Memory (Skills)	Prompts, Tool Scripts
5	/episodic_logs/	Episodic Memory (Trajectories)	Run Traces, RLVR Logs
MAP	/MAP.md & manifest.jsonl	Metacognitive Routing Index	Global Graph Pointer

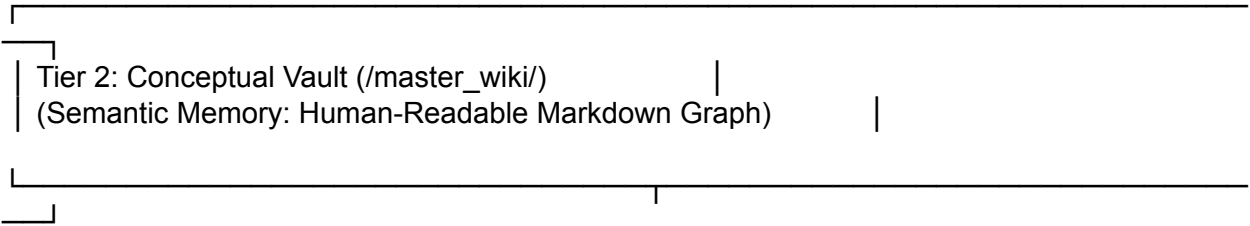
Detailed Structural Mapping

Component	System A (4-Tier)	System B (5-Tier)	Cognitive Memory Equivalence	Unified Canonical Standard	Unified Directory Path
Ground Truth Archive	Tier 1: /raw_sources/	Tier 1: /raw_pdf/	Sensory Perceptual Store (Immutable raw inputs)	Cold Archive Layer	/raw_sources/{f ormat}/ (pdf/ docx/, media/)
Synthesized Knowledge	Tier 2: /master_wiki/	Tier 2: /wiki_md/	Semantic Memory (Facts, domain ontologies)	Conceptual Knowledge Vault	/master_wiki/ (Obsidian / Markdown linked graph)
Compiled Acceleration	(Missing / Implicit)	Tier 3: /recall_jsonl/	Working Memory Extension (High-throughp ut token recall)	Machine Retrieval Cache	/recall_cache/js onl/ & /recall_cache/fa iss/
Behavioral Logic & Tools	Tier 3: /inference_skill s/	Tier 4: /repos/ & /scripts/	Procedural Memory (Skills, habits, executable routines)	Procedural Runtime & Skills	/skills/ (Prompts/YAML) + /runtimes/ (Execution code)
Trajectory & Telemetry	Tier 4: /automation_sc ripts/ (logs)	(Missing / Implicit)	Episodic Memory (Time-indexed trajectories & rewards)	Episodic Trajectory Store	/episodic_logs/ sessions/ & /episodic_logs/r lvr/
Global Routing Directory	Tier 3: "The Almanac"	Tier 5: MAP.md / Manifest.jsonl	Metacognition / Hippocampal Index	The Master Almanac	Root /MAP.md + /manifest.jsonl

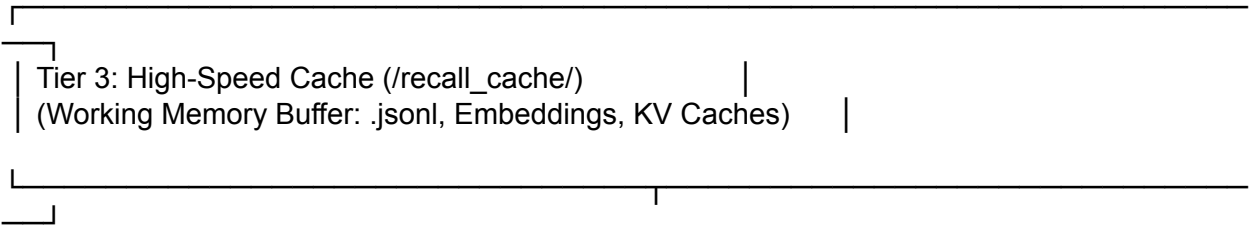
3. Cognitive-to-Engineering Architectural Pipeline



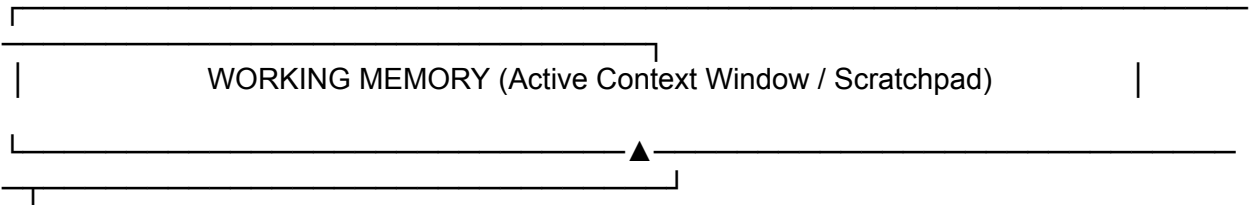
| Distill & Structure
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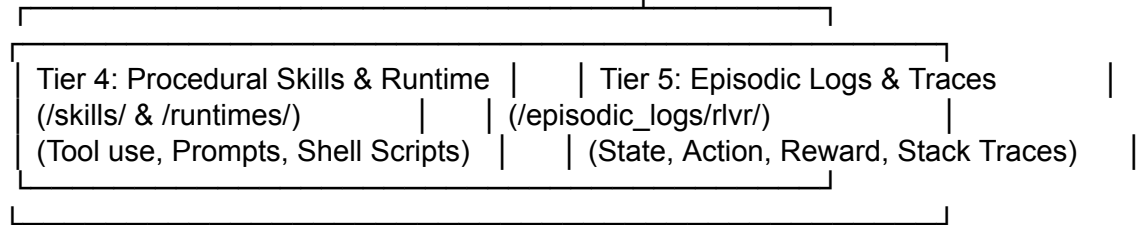
| Vectorize & Chunk
▼



| Ingest into Context Window
▼



| Queries Skill | Writes Trajectory
▼



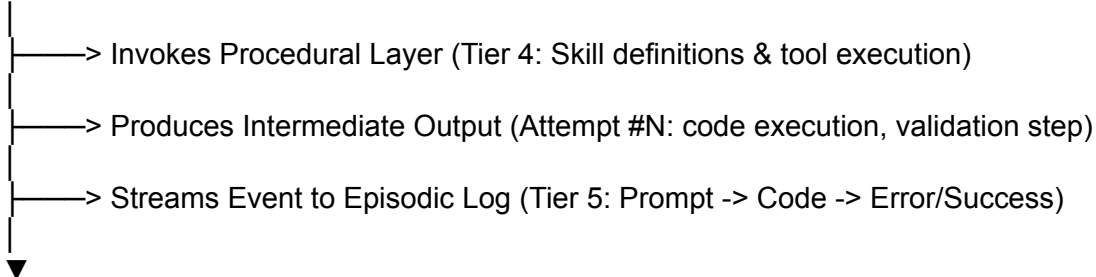
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4. The KAG Recurse & RLVR Feedback Cycle

The interaction dynamics between working context, episodic telemetry, and procedural refinement operate in a closed recursive loop:

[Attempt Start] —> Working Memory (Context Window loads task + Tier 3 context)

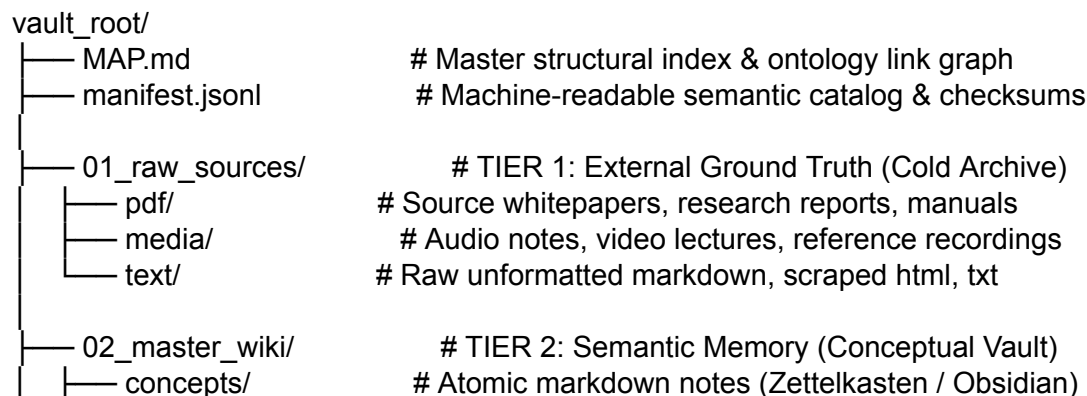


[Verifier Evaluation]

- └─ If FAILED (-1.0 Reward):
 - └─ Trigger Recursive Backtracking (load error trace into Context Window, retry)
- └─ If SUCCEEDED (+1.0 Reward):
 - └─ Commit trajectory to Episodic Vector DB
 - └─ Distill learned pattern into Tier 2 (Wiki) and Tier 4 (Procedural Prompts)

1. **Working Memory (Context Window):** Loads immediate system instructions, active variables, and dynamic tokens retrieved from **Tier 3 (/recall_cache/)**.
2. **Procedural Action (Tier 4):** The model applies algorithmic habits and tool-calling scripts from /skills/ to evaluate code, query resources, or manipulate data.
3. **Episodic Capture (Tier 5):** The raw input-output sequence of Attempt #42 is written to /episodic_logs/ as an immutable run trace (Prompt \to Action \to Environment Feedback \to Verifier Score).
4. **Reinforcement Learning via Verifiable Rewards (RLVR):** The verifier inspects the episodic record. Negative rewards trigger automated backtracking in working memory. Consistently successful runs are extracted into permanent procedural skills (Tier 4) or base policy weight updates.

5. Unified Directory Layout



— architectures/	# Technical specs and system schematics
— entities/	# Domain-specific profiles and ontologies
— 03_recall_cache/	# TIER 3: Working Memory Extension (High-Speed Retrieval)
— jsonl/	# Pre-tokenized chunks with dense metadata
— vectors/	# Vector store indices (FAISS / Chroma / HNSW)
— kv_store/	# Key-value fast recall lookups
— 04_skills_runtime/	# TIER 4: Procedural Memory (Skills & Habits)
— prompt_skills/	# Modular skill files (SKILL.md, YAML frontmatter)
— runtimes/	# Shell scripts, Python utilities, API wrappers
— policies/	# Self-correction rules and fallback schemas
— 05_episodic_logs/	# TIER 5: Episodic Memory (Historical Runs & RLVR)
— trajectories/	# Step-by-step agent traces and tool invocations
— rlvr_verifiers/	# Automated verifier scores, rewards, and feedback
— hygiene_reports/	# Linter outputs, integrity checks, drift alerts

6. Implementation Resolution Checklist

- [x] **Disambiguate "The Almanac"**: Retain MAP.md (human) and manifest.jsonl (machine) exclusively at the root level as the **Metacognitive Index**.
- [x] **Split System A Tier 4**: Separate executable automation routines into /04_skills_runtime/ and historical execution traces into /05_episodic_logs/.
- [x] **Adopt System B Tier 3**: Preserve /03_recall_cache/ (.jsonl / embeddings) to shield /02_master_wiki/ from destructive automated overwrites while enabling low-latency RAG.
- [x] **Isolate Procedural Prompts**: House agent skill templates under /04_skills_runtime/prompt_skills/ rather than bundling them into the global index.
- [x] **Align RLVR / KAG Recurse**: Treat /05_episodic_logs/ as the primary dataset generator for trajectory evaluations and long-term procedural refinement.